

**SYNOPSIS FOR MINOR PROJECT ON**  
**“Interlinked - The Urban Governance Platform”**



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## Introduction

Effective interdepartmental communication is essential for the successful execution of urban development projects. In many Indian cities, one of the recurring issues faced is the lack of coordination among government departments. For instance, newly constructed roads are often excavated again within a short period due to overlapping projects by departments such as water supply, electricity, or sewage. This not only leads to resource wastage and additional costs but also causes inconvenience to the public.

To address this problem, our project proposes an Interdepartmental Communication and Coordination Platform that will serve as a centralized portal for all government departments to share, track, and coordinate their projects across the city. The platform will feature:

- A project posting and tracking system for departments.
- A map-based visualization tool to display ongoing and upcoming projects in specific city areas.
- An anonymous public feedback portal, enhanced with AI-powered hate speech detection (using Gemini), to ensure constructive citizen participation while maintaining security.
- Restricted departmental logins, ensuring that only authorized government departments can access the portal.

### Sustainable Development Goals (SDGs) Mapping:

|                    | <b>SUSTAINABLE DEVELOPMENT GOALS<br/>(1/2/3/4/5/6/7/8/9/10/11/12/13/14/15/16/17)</b> |
|--------------------|--------------------------------------------------------------------------------------|
| <b>Interlinked</b> | 9,11,16                                                                              |

Justification:

- SDG 9: Industry, Innovation, and Infrastructure – by promoting sustainable and resilient infrastructure development through coordinated planning.
- SDG 11: Sustainable Cities and Communities – by reducing redundant work, minimizing disruption to citizens, and fostering smarter urban management.
- SDG 16: Peace, Justice, and Strong Institutions – by enabling transparent governance, interdepartmental collaboration, and public participation through feedback.

Program Outcomes (PO) Mapping:

|                    | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>Interlinked</b> | ✓   | ✓   | ✓   | ✗   | ✓   | ✓   | ✗   | ✗   | ✗   | ✗    | ✗    | ✓    |

Justification:

- PO1 (Engineering Knowledge): Apply knowledge of computer science and IT tools to design and implement a digital coordination platform.
- PO2 (Problem Analysis): Identify, formulate, and analyze the problem of inefficient interdepartmental communication and propose an effective solution.
- PO3 (Design/Development of Solutions): Design a platform integrating maps, secure logins, and AI-powered feedback handling.
- PO5 (Modern Tool Usage): Use modern technologies such as Gemini AI for natural language processing and map-based visualization tools for effective implementation.
- PO6 (The Engineer and Society): Incorporate feedback from citizens to improve accountability and ensure the project serves societal needs.
- PO12 (Life-long Learning): Encourage continuous improvement and adaptability of the system to scale for smart city requirements.

Program Specific Outcomes (PSO) Mapping:

|                    | PSO1 | PSO2 | PSO3 |
|--------------------|------|------|------|
| <b>Interlinked</b> | ✓    | ✓    | ✓    |

Justification:

- PSO1: Ability to design and develop web-based solutions integrating databases, authentication, and visualization tools.
- PSO2: Ability to apply AI/ML techniques (such as hate speech recognition) to enhance the usability and security of digital platforms.
- PSO3: Ability to create IT solutions that address real-world governance and urban planning challenges

## Problem Statement

One of the recurring challenges in urban infrastructure development is the lack of effective communication and coordination among government departments. For instance, newly constructed roads are frequently excavated again within a short span of time for activities such as laying water pipelines, electrical cables, or sewage systems. This duplication of effort leads to wasted resources, unnecessary financial expenditure, delays in project completion, and inconvenience to the public. The absence of a centralized system for interdepartmental communication remains a critical gap in current governance.

To address this issue, the proposed project, Interlinked, introduces a digital platform that streamlines interdepartmental collaboration. The platform will allow departments to post and track projects, view ongoing work on an interactive map, and coordinate with other departments to minimize redundancy. Departmental access will be secured through authorized logins, ensuring accountability and controlled participation.

In addition, Interlinked incorporates an anonymous citizen feedback portal. This feature enables the public to share suggestions or report concerns without requiring identification. To ensure that the feedback remains constructive and abuse-free, the system integrates AI-powered hate speech recognition using Gemini.

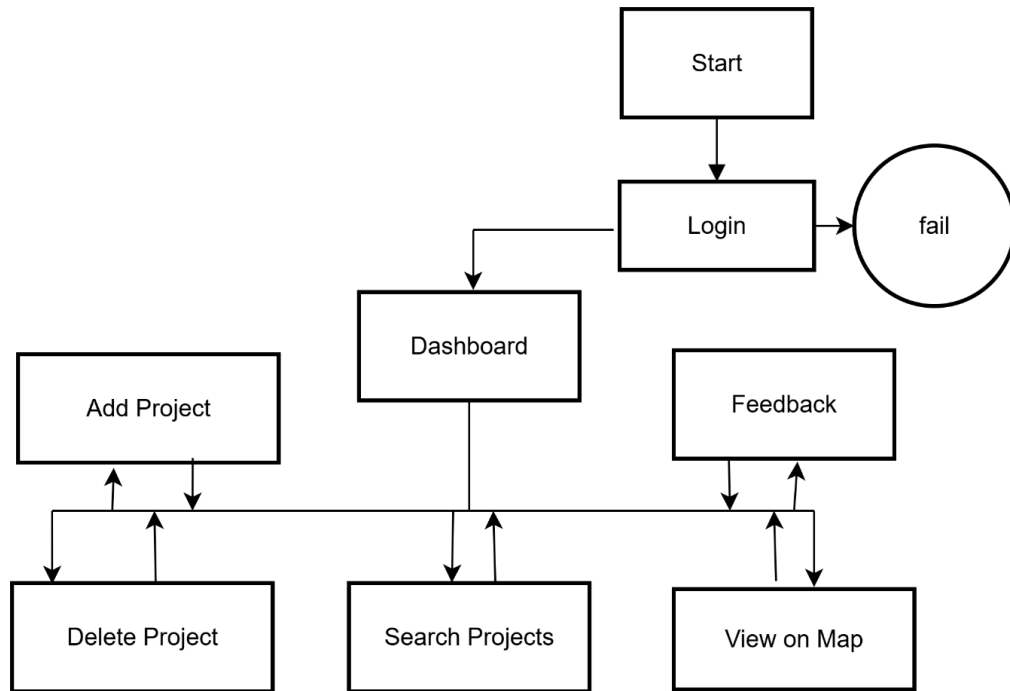
By combining project management tools, map-based visualization, and AI-enhanced public engagement, Interlinked promotes transparent governance, resource optimization, and efficient infrastructure planning. The project directly supports the UN Sustainable Development Goals (SDGs) related to sustainable infrastructure, smart cities, and strong institutions, while addressing a pressing real-world problem in urban development.

## Objectives

The primary objective of Interlinked is to build a centralized digital platform that improves coordination among government departments and enhances urban infrastructure management. The specific objectives are:

1. Facilitate Interdepartmental Collaboration – Allow departments to post, update, and track projects on a unified platform to avoid duplication of efforts.
2. Map-Based Visualization – Provide an interactive map interface that shows ongoing and upcoming projects across city regions for better planning.
3. Secure Access Control – Restrict access to authorized departmental logins to ensure accountability and prevent misuse.
4. Citizen Feedback Integration – Create an anonymous portal for public reporting and suggestions, ensuring community participation in governance.
5. AI-Powered Moderation – Implement hate speech recognition using Gemini AI to maintain the quality and safety of citizen feedback.
6. Optimize Resources and Reduce Disruptions – Prevent repeated excavation and overlapping projects, saving costs and minimizing inconvenience to the public.
7. Support Sustainable Development Goals (SDGs) – Contribute towards SDG 9, SDG 11, and SDG 16 by fostering innovation, sustainable cities, and transparent institutions.

## Block Diagram



*Figure : workflow for Interlinked*

# Requirements

## Software Requirements

- Operating System: Windows / Linux / macOS
- Frontend: React.js, Google Maps API
- Backend: Firebase
- Database: Firestore
- AI Integration: Gemini API
- Development Tools: VS Code / GitHub for version control
- Browser Support: Latest versions of Chrome, Firefox, and Edge

## Hardware Requirements

- Processor: Intel i5 or above (or equivalent)
- RAM: 8 GB minimum
- Storage: 256 GB SSD or higher
- Network: Stable high-speed internet connection

## Outcomes

The successful implementation of Interlinked will lead to the following outcomes:

1. Improved Interdepartmental Coordination
  - Government departments will be able to collaborate effectively, reducing duplication of efforts and avoiding overlapping projects.
2. Resource Optimization
  - Minimization of redundant work such as repeated excavation will save financial resources, time, and manpower.
3. Enhanced Transparency
  - A centralized platform with map-based visualization will make project planning and execution more transparent and accessible to authorized departments.
4. Citizen Participation
  - The anonymous feedback portal will provide citizens with a voice in infrastructure projects, strengthening the bond between public institutions and society.
5. Safe and Constructive Feedback Mechanism
  - With Gemini AI-powered hate speech detection, feedback will remain meaningful, free from abuse, and beneficial for decision-making.
6. Public Convenience
  - Reduced disruptions in public spaces (e.g., repeated road digging) will improve the quality of life for citizens.
7. Support to Sustainable Development Goals (SDGs)
  - Contributes directly to SDG 9 (Innovation and Infrastructure), SDG 11 (Sustainable Cities), and SDG 16 (Strong Institutions), aligning with global sustainability practices.

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